Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, SQL, Java, C/C++, JavaScript, R
Business Analysis & Analytics: Requirements Analysis, Exploratory Data Analysis (EDA), Ad-Hoc Reporting, KPI Tracking, Trend Analysis, Statistical Analysis, Business Intelligence, Data-Driven Decision Making
Data Visualization & BI: Power BI, Microsoft Excel (Pivot Tables, Lookups, Charts), Dashboard Development, Data Visualization
Data Science & Machine Learning: Scikit-learn, Machine Learning, Time-Series Forecasting, Anomaly Detection, Deep Learning (PyTorch), Probabilistic Modeling (BG/NBD, Gamma-Gamma), Model Evaluation
Databases: MySQL, SQLite, Redis (NoSQL), SQL Query Optimization
Development & Tools: Flask, FastAPI, React.js, Node.js, REST APIs, Git, GitHub, GitHub Actions, Docker, PyTest, CI/CD

EXPERIENCE

VOIS – Vodafone Idea Foundation* |LINK** — ***Aug – Sep, 2025

- Analyzed 2 real-world business datasets (8,500+ records) using AI-assisted analytics to identify trends, generate 15 actionable business insights, and support data-driven decision-making.
- Validated analytical findings for accuracy and consistency, ensuring reliable recommendations for business planning and operational improvements.
- Designed a standardized analytics workflow for data preprocessing, insight generation, and validation, enabling efficient conversion of raw data into meaningful business recommendations.

PROJECTS

***RedisLite* |LINK**

- Engineered a Redis-inspired in-memory data store in Python with a custom RESP protocol parser, multithreaded TCP server, and an O(1) LRU cache with lazy/active TTL expiration — core data pipeline infrastructure for high-throughput storage and retrieval.
- Designed a four-layer architecture (transport, storage engine, cache/TTL, persistence) implementing Write-Ahead Logging (WAL) and automatic crash recovery to ensure data integrity.
- Validated the system with 46 automated PyTest unit tests, benchmarking 619K SET ops/sec in-memory and 2.8K ops/sec with durable WAL persistence.

***Real-Time Stock Market Anomaly Detector* |LINK**

- Performed exploratory data analysis on NSE stock market data to identify trends and abnormal patterns, combining statistical (Isolation Forest) and deep learning (LSTM Autoencoder) techniques.
- Ran ad-hoc analysis on financial news sentiment (FinBERT) as a secondary signal to explain anomalies, translating unstructured data into structured, actionable insight.
- Achieved ~88% precision and ~85% recall on a labeled validation set, evaluating trade-offs between detection sensitivity and false positives.

***Customer Lifetime Value Engine* |LINK**

- Built a probabilistic model (BG/NBD + Gamma-Gamma) on 1M+ historical transaction records to forecast customer lifetime value directly analogous to modeling player value and monetization potential.
- Segmented customers into value-based cohorts to identify which segments drive the most revenue, supporting prioritization decisions.
- Identified the top 20% of customers driving ~65% of forecasted revenue, visualized through Power BI dashboards for stakeholder-facing reporting.

***AI-Powered Expense Receipt Scanner & Analytics Platform* |LINK**

- Built an analytics dashboard tracking spend trends, budget adherence, and category-wise insights — directly analogous to tracking user engagement/monetization metrics for a product.
- Automated data extraction and processing from unstructured receipt data using EasyOCR and NLP-based parsing, improving processing efficiency by 40%.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional |**LINK** 2026
- Deloitte Australia Data Analytics Job Simulation |**LINK** 2025
- AWS APAC Solutions Architecture virtual experience program | **LINK** 2025